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//Dice Roller

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AI Tip 1. Research how to calculate the exact dimensions of the array.

Answer: I didn't use AI on this question, I write this code on my own:

```
“ mySumRolls = new int[this.myNumberOfSides*this.myNumberOfDice-  
this.myNumberOfDice+1]”
```

AI Tip 2. Research how to clone an array in Java

Answer: I searched this question in YouTube, and write this:

```
public int[] results() {  
    if(mySumRolls==null) {  
        return null;  
    }  
    else  
  
        return mySumRolls.clone();  
}
```

Resource: <https://www.youtube.com/watch?v=6uemeqwFNeQ>

AI Tip 3. Inspect the Integer class for a constant that is useful in the cases where we don't care about the maximum value.

Answer: I asked Gemini: in java, how to define positive int infinite.

Gemini:

The `int` primitive type in Java (and all other integer primitive types like `byte`, `short`, and `long`) does not have a special value to represent positive infinity.

This is because `int` is a **fixed-size, two's complement integer type**. It can only hold a whole number within a defined, finite range.

Key Points for Integers

1. **Fixed Range:** A standard Java `int` is a 32-bit value. Its maximum value is $2^{31} - 1$, which is represented by the constant `Integer.MAX_VALUE` (**2,147,483,647**).
2. **No Infinity Constant:** Unlike `double` or `float` (which follow the IEEE 754 standard for floating-point arithmetic and include special bit patterns for infinity), the `Integer` wrapper class and the `int` primitive do not have a concept of "infinity."
3. **Overflow Behavior:** When an arithmetic operation on integers exceeds `Integer.MAX_VALUE`, it does not become "infinity." Instead, it **wraps around** to the most negative value (`Integer.MIN_VALUE`).

How to Handle "Infinity" with `int`

Since true infinity is not possible, developers typically use one of two methods to functionally represent "positive infinity" in an integer context:

1. **Use `Integer.MAX_VALUE`:**

This is the most common practice. You treat the absolute maximum possible integer value as a stand-in for infinity in comparisons and logic.

So I go with: `Integer.MAX_VALUE`

Here are other questions that I asked AI:

1. In Java, how to print an array to string .

Answer:



You can print an array to a string in Java using the `Arrays.toString()` method, or for multi-dimensional arrays, the `Arrays.deepToString()` method.

Here's how to use both:

1. Using `Arrays.toString()` (For Single-Dimensional Arrays)

This is the standard way for one-dimensional arrays of any primitive type or object.

Java



```
import java.util.Arrays;

public class ArrayToStringExample {
    public static void main(String[] args) {
        // 1. Primitive Array (int)
        int[] intArray = {10, 20, 30, 40, 50};
        String intString = Arrays.toString(intArray);
        System.out.println("Int Array String: " + intString);
        // Output: Int Array String: [10, 20, 30, 40, 50]

        // 2. Object Array (String)
        String[] stringArray = {"apple", "banana", "cherry"};
        String stringArrayString = Arrays.toString(stringArray);
        System.out.println("String Array String: " + stringArrayString);
        // Output: String Array String: [apple, banana, cherry]
    }
}
```

I go with first answer “`Arrays.toString()`”.

2. How can I save `printf` into a string variable?

Answer:

```
java System.out.printf("%6d)%7d%n",i,mySumRolls[i-  
this.myNumberOfDice]);
```

what if i want to save all in the printf into a variable called res ?

Good question 👍

In Java, `System.out.printf()` **prints directly to the console** and does **not return** a string — it returns a `PrintStream` object, so you can't capture its output directly into a variable.

If you want to **save the formatted text** (the same as what `printf` would print) into a variable called `res`, use `String.format()` instead:

java

 Copy code

```
String res = String.format("%6d)%7d%n", i, mySumRolls[i - this.myNumberOfDice]);
```

✅ `String.format()` works exactly like `printf`, but instead of printing, it **returns** the formatted text as a string.

That is all! Thank you!